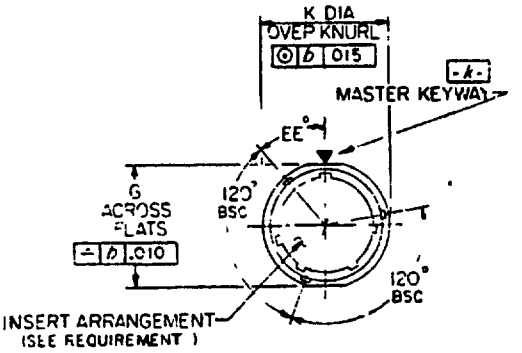
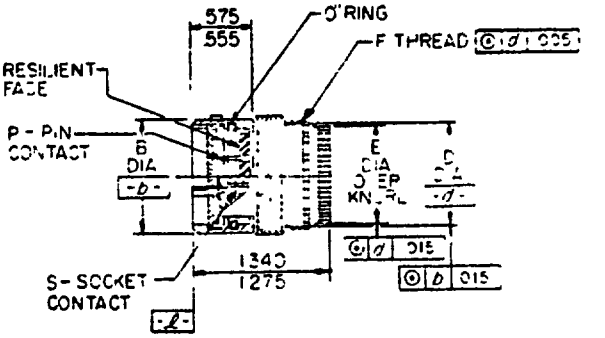
MILITARY SPECIFICATION SHEET

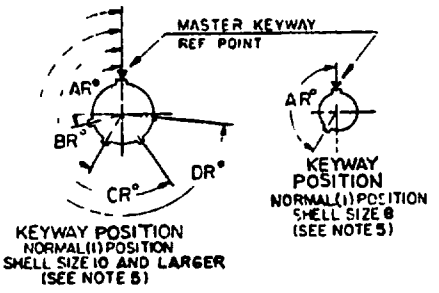
CONNECTORS, ELECTRICAL, CIRCULAR, HIGH DENSITY, QUICK DISCONNECT, ENVIRONMENT RESISTING; AND ACCESSORIES: RECEPTACLE, CABLE CONNECTING, CRIMP-TYPE CONTACTS. CLASS A, F AND E (SERIES 2)

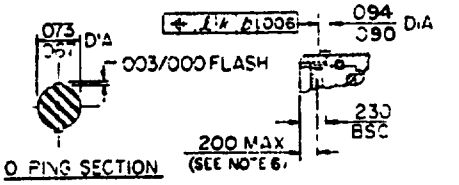
The complete requirements for procuring the connector described herein shall consist of this document and the latest issue of Specification MIL-C-81511.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.









INCHES	MM	INCHES	MM
.003	.08	.200	5.08
.005	.13	.230	5.84
.006	.15	.555	14.10
.010	.25	.575	14.61
.015	.38	1.275	32.39
.067	1.70	1.340	34.04
.073	1.85		
.090	2.29		
.094	2.39		

FSC 5935

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TABLE I.

Shell size	B dia	D dia	E dia	F thread	G across flats	K dia	AR° BSC	BR° BSC	CR° BSC	DR° BSC	EE° BSC Position 1	2	3	4	5	6
8	.553 (14.05) .547 (13.89)	.430 (10.92) .424 (10.77)	.422 (10.72) .407 (10.34)	½-28UNEF-2A	.565 (14.35) .553 (14.05)	.780 (19.81) .760 (19.30)	128				40	40	40	40	40	40
10	.678 (17.22) .672 (17.07)	.555 (14.10) .549 (13 94)	.548 (13.92) .533 (13.54)	5/8-28UN-2A	.690 (17.53) .678 (17.22)	.900 (22.86) .880 (22.35)	105	140	215	265	60	46	46	60	89	60
14	.928 (23.57) .922 (23.42)	.805 (20.45) .799 (20.29)	.798 (20.27) .783 (19.89)	7/8-28UN-2A	.940 (23.88) .928 (23.57)	1.150 (29.21) 1.130 (28.70)	105	140	215	265	60	46	46	60	89	60
16	1.054 (26.77) 1.048 (26.62)	.930 (23.62) .924 (23.47)	.924 (23.47) .909 (23.09)	1-28UN 2A	1.066 (27.08) 1.054 (26.77)	1,280 (32.51) 1.260 (32.00)	105	140	215	265	50	50	50	50	50	50
18	1.178 (29.92) 1.172 (29.77)	1.055 (26.80) 1.049 (26.64)	1.048 (26.62) 1.033 (26.24)	1-1/8-28UN-2A	1.190 (30.23) 1.178 (29.92)	1.400 (35.56) 1.380 (35.05)	105	140	215	265	50	50	50	50	50	50

NOTES
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Dimensions are after plating.
5. Normal mating keyway position shown, for alternate keyway positions see MIL-C-81511, figure 4.
6. Distance from front of shell to the point at which a gage pin, having the same basic diameter as plug shell, engages the spring member.
7. For shell sizes 20, 22 and 24 use MIL-C-81511/25 connectors

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REQUIREMENTS:

Dimensions and configuration: See figure and table I.

Series 2

Classes: A, F, and E

Shell size: See table II.

Insert arrangement. See supplement for MS sheet corresponding to desired shell size.

Contacts: Style P or S, crimp type.

Shell polarization: 1 thru 6, key position.

Weight: See table II.

Marking. Part number on shell.

Mating counterpart: Plug connectors specified in MIL-C-81511/6 and /34.

Insertion and removal tool: As specified in MS3323, MS3342, and MS3344.

Part number: The part number shall consist of M81511/05 together with the applicable codes for class, shell size, insert arrangement, contact style, and key positions Information concerning the formulation of the part number can be found in MIL-C-81511.

TABLE II.

Shell size	Weight (REF) 1/ (pounds)
18 (code F)	.074
16 (code E)	066
14 (code D)	.052
10 (code B)	.031
8 (code A)	.024

1/ Removable crimp contacts not included in weight.
(See contact drawings for weight of pin and socket contact.)

Custodians:
Army - EL
Navy - AS
Air Force - 17

Preparing activity:
Navy - AS

Review activities:
Army - MI, MU, AV, WC
Navy - SH, EC, OS
Air Force - 11, 80
DSA - ES

(Project No. 5935-1732-5)

User activities:
Army - ME, AT
Navy - MC
Air Force -

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS. This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-81511/5E - CONNECTORS, ELECTRICAL, CIRCULAR, HIGH DENSITY, QUICK DISCONNECT, ENVIRONMENT RESISTING; AND ACCESSORIES: RECEPTACLE, CABLE CONNECTING, CRIMP-TYPE CONTACTS, CLASS A, F AND E (SERIES 2)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 86 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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DEPARTMENT OF THE NAVY

Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Weapons Engineering Standardization Office (Code X)
Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

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